

First-Principles Derivation of the Steiner Sentence Length Distribution

Speculative Results Series, Paper 67

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Abstract

We prove that Steiner sentences in the Syracuse graph have length distribution $P(\text{sentence length} = k) = 3^{k-1}/4^k$ for all $k \geq 1$. The proof proceeds in three steps. First, we establish the exact Syracuse transition matrix on residue classes modulo 8 from arithmetic alone: the mod-8 map is uniform from classes 1 and 5, while classes 3 and 7 have restricted (forbidden) transitions determined by mod-16 arithmetic. Second, we identify the invariant $d_3 = d_7$ — equal weight on classes 3 and 7 in the surviving (non-terminated) distribution — and show it is preserved by the transition matrix unconditionally, for any value of d_1 . Third, we show that the invariant $d_3 = d_7$ is precisely the algebraic condition under which the surviving mass decays by exactly 3/4 per step, giving $P(k) = 3^{k-1}/4^k$.

The proof requires no ergodic theory and no appeal to the Collatz conjecture. It is a direct consequence of two classical arithmetic facts: that $3n + 1 \equiv 2 \pmod{4}$ for $n \equiv 3$ or $7 \pmod{8}$ (mod-8 forbidden transitions), and that multiplication by 3 is a bijection on $\mathbb{Z}/2^j\mathbb{Z}$ (uniform distribution from classes 1 and 5).

Appendix A provides a complete empirical validation: 10^5 sentences sampled uniformly over $[1, 10^{15}]$, the sentence-start distribution, the within-sentence word entry distribution, the sentence length table comparing the naive $(1/2)^k$ prediction and the theoretical $3^{k-1}/4^k$, and χ^2 goodness-of-fit tests. The naive prediction is rejected with $\chi^2 \approx 10^6$; the theoretical formula is consistent ($p \approx 0.61$).

Status. This paper contains proved theorems and complete empirical validation. All theoretical results are unconditional.

Contents

1	Background and Notation	2
2	The Syracuse Transition Matrix mod 8	2

3	The Sentence as a First-Passage Problem	3
4	The Invariant $d_3 = d_7$	4
5	Main Theorem	5
6	Discussion	6
6.1	The role of class 5	6
6.2	Connection to Paper 66	6
6.3	Why $3^{k-1}/4^k$ rather than $(1/2)^k$	7
A	Empirical Validation	7

1 Background and Notation

We work throughout with odd positive integers.

Definition 1.1 (Syracuse map). The *Syracuse map* $S : \mathbb{O} \rightarrow \mathbb{O}$ on odd positive integers is

$$S(n) = \frac{3n + 1}{2^{v_2(3n+1)}},$$

where $v_2(m)$ denotes the 2-adic valuation of m .

Definition 1.2 (Steiner sentence). A *Steiner sentence* is a maximal sequence of consecutive applications of S from an initial node $b_0 = S(n)$ with $n \equiv 5 \pmod{8}$, terminating at the first node $b_k \equiv 5 \pmod{8}$. The *sentence length* is the number of steps k .

Remark 1.3. Steiner sentences are the fundamental composite building blocks of Collatz paths; see Paper 33 [1] for the full Steiner circuit framework. The sentence start $b_0 = S(n)$ is the node immediately following a $5 \pmod{8}$ node in the orbit, and the sentence ends at the next $5 \pmod{8}$ node.

Paper 65 [2] was the first attempt at this subject. It observed empirically that $P(\text{sentence length} = k) = \frac{1}{3}(3/4)^k$, equivalently $3^{k-1}/4^k$, and stated the result as a conjecture. The present paper supersedes Paper 65 by providing a complete proof, and incorporates its experimental record in Appendix A.

2 The Syracuse Transition Matrix mod 8

We model the sentence as a Markov chain on residue classes modulo 8. The transition probability $P[a][b]$ is the fraction of odd integers $n \equiv a \pmod{8}$ for which $S(n) \equiv b \pmod{8}$, measured over the natural density.

Theorem 2.1 (Syracuse transition matrix mod 8). *The transition matrix on $\{1, 3, 5, 7\}$ (mod 8) is:*

$$P = \begin{array}{c|cccc} & 1 & 3 & 5 & 7 \\ \hline 1 & 1/4 & 1/4 & 1/4 & 1/4 \\ 3 & 1/2 & 0 & 1/2 & 0 \\ 5 & 1/4 & 1/4 & 1/4 & 1/4 \\ 7 & 0 & 1/2 & 0 & 1/2 \end{array}$$

(rows = source class, columns = target class).

Proof. We prove each row by direct arithmetic.

Rows 1 and 5 (uniform rows). Write $n = 8k + r$ for $r \in \{1, 5\}$. Then $3n + 1 = 24k + (3r + 1)$, so $3n + 1 \pmod{8}$ depends only on r modulo 8, not on k . For $r = 1$: $3(1) + 1 = 4$, so $3n + 1 \equiv 4 \pmod{8}$, giving $v_2(3n + 1) \geq 2$. Specifically, $S(n) = (3n + 1)/4$ and the residue of $S(n) \pmod{8}$ depends on k . Since $\gcd(3, 2^j) = 1$, multiplication by 3 is a bijection on $\mathbb{Z}/2^j\mathbb{Z}$ for every $j \geq 1$. Therefore as k ranges uniformly over a complete residue system mod 2^j , the value $3n + 1 \pmod{2^j}$ is uniform over $2^j\mathbb{Z}$. In particular $S(n) \pmod{8}$ is uniform over $\{1, 3, 5, 7\}$, giving $P[1][b] = 1/4$ for each $b \in \{1, 3, 5, 7\}$. The same argument applies to $r = 5$ (where $3n + 1 = 8(3k + 2)$ and uniformity of $3k + 2 \pmod{2^j}$ follows from $\gcd(3, 2^j) = 1$).

Row 3 (restricted to $\{1, 5\}$). Write $n = 8k + 3$. Then $3n + 1 = 24k + 10 = 2(12k + 5)$. Since $12k$ is even, $12k + 5$ is always odd, so $v_2(3n + 1) = 1$ exactly and $S(n) = 12k + 5$. Now $12k + 5 \pmod{8}$: $12k \equiv 4k \pmod{8}$, so $12k + 5 \equiv 4k + 5 \pmod{8}$. For k odd: $4k \equiv 4 \pmod{8}$, giving $12k + 5 \equiv 1 \pmod{8}$. For k even: $4k \equiv 0 \pmod{8}$, giving $12k + 5 \equiv 5 \pmod{8}$. Since k is odd and even with equal density, $P[3][1] = P[3][5] = 1/2$ and $P[3][3] = P[3][7] = 0$.

Row 7 (restricted to $\{3, 7\}$). Write $n = 8k + 7$. Then $3n + 1 = 24k + 22 = 2(12k + 11)$. Since $12k + 11$ is always odd, $v_2(3n + 1) = 1$ exactly and $S(n) = 12k + 11$. Now $12k + 11 \pmod{8}$: $12k + 11 \equiv 4k + 3 \pmod{8}$. For k odd: $4k \equiv 4$, giving $4k + 3 \equiv 7 \pmod{8}$. For k even: $4k \equiv 0$, giving $4k + 3 \equiv 3 \pmod{8}$. Therefore $P[7][3] = P[7][7] = 1/2$ and $P[7][1] = P[7][5] = 0$. $\square$

Remark 2.2 (Forbidden transitions). The zero entries in P — transitions $3 \rightarrow 3$, $3 \rightarrow 7$, $7 \rightarrow 1$, $7 \rightarrow 5$ — are *structurally forbidden* for all n , not just on average. Classes 3 and 7 each map into a strict subset of $\{1, 3, 5, 7\}$. This structural restriction is the engine of the proof.

3 The Sentence as a First-Passage Problem

A sentence is the first-passage time of the Markov chain (P, π_0) to state 5, where π_0 is the sentence-start distribution.

Lemma 3.1 (Sentence-start distribution). *The sentence-start distribution is $\pi_0 = (1/4, 1/4, 1/4, 1/4)$ — uniform on $\{1, 3, 5, 7\}$.*

Proof. The sentence starts at $b_0 = S(n)$ with $n \equiv 5 \pmod{8}$. By Theorem 2.1, row 5 of P is uniform, so $b_0 \pmod{8}$ is uniform on $\{1, 3, 5, 7\}$. $\square$

The sentence terminates at the first step where the chain visits state 5. The sentence length k is this first-passage time.

Definition 3.2 (Surviving distribution). Let $d_s^{(k)}$ denote the probability that the chain is at state s after k steps without having visited state 5. Formally, $d_s^{(0)} = \pi_0(s)$ and for $k \geq 0$:

$$d_s^{(k+1)} = \sum_{t \neq 5} d_t^{(k)} P[t][s], \quad s \neq 5.$$

The *surviving mass* is $M_k = \sum_{s \neq 5} d_s^{(k)}$.

Then:

$$P(\text{sentence length} = k) = M_{k-1} - M_k = \sum_s d_s^{(k-1)} P[s][5].$$

4 The Invariant $d_3 = d_7$

The key structural fact is a symmetry between classes 3 and 7 in the surviving distribution.

Lemma 4.1 (Invariant). $d_3^{(k)} = d_7^{(k)}$ for all $k \geq 0$.

Proof. By Definition 3.2, the surviving distribution sums only over sources $t \neq 5$; class 5 contributes nothing because once the chain reaches 5 the sentence terminates (the mass is absorbed, not propagated). The transition update for states 3 and 7 is therefore:

$$\begin{aligned} d_3^{(k+1)} &= d_1^{(k)} \cdot P[1][3] + d_3^{(k)} \cdot P[3][3] + d_7^{(k)} \cdot P[7][3] \\ &= \frac{1}{4} d_1^{(k)} + 0 \cdot d_3^{(k)} + \frac{1}{2} d_7^{(k)}, \\ d_7^{(k+1)} &= d_1^{(k)} \cdot P[1][7] + d_3^{(k)} \cdot P[3][7] + d_7^{(k)} \cdot P[7][7] \\ &= \frac{1}{4} d_1^{(k)} + 0 \cdot d_3^{(k)} + \frac{1}{2} d_7^{(k)}. \end{aligned}$$

The two expressions are identical: $d_3^{(k+1)} = d_7^{(k+1)}$ for any $d_1^{(k)}, d_3^{(k)}, d_7^{(k)}$. By induction, since $d_3^{(0)} = d_7^{(0)} = 1/4$ (Lemma 3.1), $d_3^{(k)} = d_7^{(k)}$ for all $k \geq 0$. $\square$

Remark 4.2 (Structural source of the invariant). The invariant holds because columns 3 and 7 of P , restricted to rows $\{1, 3, 7\}$, are *identical*:

$$P[\cdot][3]|_{\{1,3,7\}} = P[\cdot][7]|_{\{1,3,7\}} = \left(\frac{1}{4}, 0, \frac{1}{2}\right).$$

This column symmetry is a direct consequence of Theorem 2.1:

- *From class 1*: uniform output gives equal weight $1/4$ to both 3 and 7.
- *From class 3*: forbidden to reach 3 or 7 (both have weight 0).
- *From class 7*: reaches 3 and 7 with equal probability $1/2$ each.

The invariant is unconditional — it holds for *any* value of d_1 , not just at the fixed point of the chain.

5 Main Theorem

Theorem 5.1 (Sentence length distribution). *For any $k \geq 1$:*

$$P(\text{sentence length} = k) = \frac{3^{k-1}}{4^k}.$$

Equivalently, $P(\text{sentence length} = k) = \frac{1}{3} \left(\frac{3}{4}\right)^k$.

Proof. We use the splitting: $P(\text{sentence length} = 1) = \pi_0(5) = 1/4$ (sentences that start at $b_0 \equiv 5$ terminate immediately). For $k \geq 2$, the sentence does not terminate at step 1 (probability $3/4$) and then terminates at step $k-1$ from the conditional start, i.e. $P(\text{length} = k) = \frac{3}{4} \cdot g_{k-1}$, where g_j is the first-passage probability to 5 starting from the conditional distribution $\pi_0(\cdot \mid \cdot \neq 5) = (1/3, 1/3, 1/3)$ uniform on $\{1, 3, 7\}$.

It suffices to show $g_j = (3/4)^{j-1}/4$ for all $j \geq 1$, which follows if the surviving mass M_j starting from uniform $\{1, 3, 7\}$ satisfies $M_j = (3/4)^j$.

Step 1: $d_3 = d_7$ is preserved. The conditional start has $d_3^{(0)} = d_7^{(0)} = 1/3$, so Lemma 4.1 applies: $d_3^{(k)} = d_7^{(k)}$ for all $k \geq 0$.

Step 2: Surviving mass decays by exactly $3/4$. From Definition 3.2, $M_{k+1} = \sum_{s \neq 5} d_s^{(k+1)} = \sum_{s \neq 5} \sum_{t \neq 5} d_t^{(k)} P[t][s] = \sum_{t \neq 5} d_t^{(k)} \sum_{s \neq 5} P[t][s] = \sum_{t \neq 5} d_t^{(k)} (1 - P[t][5])$, giving:

$$M_{k+1} = d_1^{(k)} (1 - P[1][5]) + d_3^{(k)} (1 - P[3][5]) + d_7^{(k)} (1 - P[7][5]).$$

Substituting $P[1][5] = 1/4$, $P[3][5] = 1/2$, $P[7][5] = 0$:

$$M_{k+1} = \frac{3}{4} d_1^{(k)} + \frac{1}{2} d_3^{(k)} + d_7^{(k)}.$$

We claim $M_{k+1} = \frac{3}{4} M_k = \frac{3}{4} (d_1^{(k)} + d_3^{(k)} + d_7^{(k)})$. This requires:

$$\frac{3}{4} d_1^{(k)} + \frac{1}{2} d_3^{(k)} + d_7^{(k)} = \frac{3}{4} d_1^{(k)} + \frac{3}{4} d_3^{(k)} + \frac{3}{4} d_7^{(k)},$$

which simplifies to $\frac{1}{2} d_3^{(k)} + d_7^{(k)} = \frac{3}{4} (d_3^{(k)} + d_7^{(k)})$, i.e. $d_7^{(k)} = d_3^{(k)}$. This is exactly Lemma 4.1.

Step 3: Initial condition. At $k = 0$, $M_0 = 1$ (the full conditional mass on $\{1, 3, 7\}$). By induction, $M_k = (3/4)^k$ for all $k \geq 0$.

Conclusion. $g_j = M_{j-1} - M_j = (3/4)^{j-1} - (3/4)^j = (3/4)^{j-1} (1/4) = 3^{j-1}/4^j$. Therefore:

$$P(\text{length} = 1) = \frac{1}{4} = \frac{3^0}{4^1},$$

$$P(\text{length} = k) = \frac{3}{4} \cdot g_{k-1} = \frac{3}{4} \cdot \frac{3^{k-2}}{4^{k-1}} = \frac{3^{k-1}}{4^k}, \quad k \geq 2.$$

Both cases give $3^{k-1}/4^k$, completing the proof. □

Corollary 5.2 (Normalisation). $\sum_{k=1}^{\infty} P(\text{sentence length} = k) = 1$.

Proof. $\sum_{k=1}^{\infty} \frac{3^{k-1}}{4^k} = \frac{1}{4} \sum_{k=1}^{\infty} \left(\frac{3}{4}\right)^{k-1} = \frac{1}{4} \cdot \frac{1}{1-3/4} = 1$. □

6 Discussion

6.1 The role of class 5

The formula $3^{k-1}/4^k$ is the geometric distribution with success probability $1/4$, as if at each step the chain draws uniformly from $\{1, 3, 5, 7\}$ and terminates on drawing 5. This appearance is striking because class 5 plays three distinct and mutually reinforcing roles in the sentence structure.

Class 5 as the termination state. A sentence terminates exactly when it reaches a node $\equiv 5 \pmod{8}$. Since class 5 is the unique absorbing boundary of the chain (once the chain visits class 5, the sentence ends), it is structurally absent from all intermediate steps. Empirically, class 5 accounts for only $\approx 6.2\%$ of within-sentence word entries (Appendix A, Table 2) — one entry per sentence, uniformly distributed across sentence lengths — rather than the 25% it would contribute if words were unrestricted.

Class 5 as the uniform source. The sentence starts at $b_0 = S(n)$ with $n \equiv 5 \pmod{8}$. By row 5 of the transition matrix (Theorem 2.1), class 5 maps uniformly over $\{1, 3, 5, 7\}$: any of the four classes is equally likely as the first word. In particular, exactly $1/4$ of sentences have $b_0 \equiv 5$, meaning length $k = 1$. This is the correct $P(k = 1) = 1/4$, not the naive $1/2$. Class 5 is the unique class for which $v_2(3n + 1) \geq 3$ always (proved in Paper 66 [3]), which forces the extra halvings that spread the output uniformly.

Class 5 as the structural asymmetry. The transition $7 \rightarrow 5$ is structurally forbidden: $P[7][5] = 0$. Class 5 is therefore unreachable in one step from class 7, while it is reachable from classes 1 (with probability $1/4$) and 3 (with probability $1/2$). This asymmetry — class 5 reachable from some classes but not others — is precisely what forces the invariant $d_3 = d_7$: nodes in class 7 can only reach class 5 via a detour through class 3 or class 7, never directly. The proof in Section 4 shows that this forced detour maintains equal weight on classes 3 and 7 throughout the chain, and Section 5 shows that this equal weight is the algebraic condition that makes the effective termination probability exactly $1/4$ at every step, regardless of the current value of d_1 .

6.2 Connection to Paper 66

Paper 66 [3] proves that $E[v_2(3n+1) \mid n \equiv r \pmod{8}]$ equals exactly 2, 1, 4, 1 for $r = 1, 3, 5, 7$ respectively. The present paper shows that the $3/4$ decay rate of sentences is a direct consequence of the same arithmetic:

- The $1/4$ probability of immediate termination ($b_0 \equiv 5$) comes from the uniform output of class 5 (Paper 66, class 5 is the unique class with $c_r \geq 3$, causing its output to be uniform over all four classes).
- The $1/2$ termination probability from class 3 comes from the mod-16 split proved in Theorem 2.1: exactly half of class 3 nodes map to class 5.
- The invariant $d_3 = d_7$ — and hence the geometric decay — follows from the structural symmetry between the forbidden transitions of classes 3 and 7.

6.3 Why $3^{k-1}/4^k$ rather than $(1/2)^k$

The naive prediction $(1/2)^k$ arises from the “3-node coin flip” model: treat each word in the sentence as independently terminating whenever it visits the 3-node, with the 3-node split giving probability $1/2$. Under this model only classes 3 and 7 can terminate, and if the within-sentence distribution were uniform over $\{1, 3, 5, 7\}$, the effective per-step termination probability would be:

$$\frac{1}{4} \cdot 0 + \frac{1}{4} \cdot \frac{1}{2} + \frac{1}{4} \cdot 1 + \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{2},$$

where the terms correspond to classes 1 (never terminates via 3-node), 3 (terminates with prob $1/2$), 5 (terminal by definition), and 7 (terminates with prob $1/2$). This model is wrong in two ways. First, class 1 *can* reach class 5 directly: $P[1][5] = 1/4$, so the naive assignment of 0 to class 1 understates termination from class 1. Second and more importantly, class 7 *cannot* reach class 5 at all: $P[7][5] = 0$, so the naive assignment of $1/2$ to class 7 overstates it. The net effect of these two corrections is a reduction in the effective termination rate from $1/2$ to $1/4$. The structural reason is the forbidden transition $7 \rightarrow 5$: a chain in class 7 must first transit to class 3 or 7 (remaining in the growth classes), where it may then reach class 5 on a subsequent step. The invariant $d_3 = d_7$ (Lemma 4.1) shows that these two errors cancel exactly and yield an effective rate of precisely $1/4$, independent of d_1 .

Acknowledgements

This paper is part of the speculative results stream (64+k) of the Collatz programme. The proof structure was found through numerical exploration in `notebooks/steiner-branch-model.ipynb`.

A Empirical Validation

This appendix records the complete numerical experiment establishing the sentence length distribution. The experiment was first reported in Paper 65 [2] and is reproduced here in full as the empirical record accompanying this paper’s proof.

Sampling Protocol

1. Draw t uniformly at random from $[1, T_{\max}]$ with $T_{\max} = 10^{15}$.
2. Set $n = 8t + 5$ (so $n \equiv 5 \pmod{8}$).
3. Compute $b_0 = S(n)$, the first Steiner word of the new sentence.
4. Follow consecutive Steiner circuits from b_0 , counting words, until the first output $b \equiv 5 \pmod{8}$. Record the word count as k .

$N = 10^5$ independent sentences were collected with seed 42. Independence between samples is ensured by the random draw of t ; no orbit tails are shared.

Sentence-start Distribution

By Theorem 2.1, row 5 of P is uniform, so $b_0 = S(n) \bmod 8$ should be uniform on $\{1, 3, 5, 7\}$.

$b_0 \bmod 8$	Count	Observed fraction	Expected (uniform)
1	25181	0.25181	0.25000
3	25023	0.25023	0.25000
5	24852	0.24852	0.25000
7	24944	0.24944	0.25000

Table 1: Sentence-start distribution: $b_0 = S(n)$, $n \equiv 5 \pmod{8}$, $N = 10^5$ samples. The distribution is uniform on $\{1, 3, 5, 7\}$ as predicted by Theorem 2.1.

Within-sentence Steiner Word Entry Distribution

The entry class of each Steiner word within a sentence determines its termination behaviour. Class 5 words always terminate the sentence, so they appear only as the final word — their within-sentence frequency is structurally suppressed to approximately $1/(k+1)$ for a sentence of length k , averaging to $\approx 1/16$ empirically.

Entry mod8	Count	Fraction	Visits 3-node?	Term. prob.
1	125450	0.31333	no	0
3	125119	0.31250	yes	1/2
5	24852	0.06207	no	1 (terminal only)
7	124955	0.31209	yes	1/2
Total	400376			

Table 2: Within-sentence Steiner word entry distribution across all 10^5 sentences (400376 total words). Class 5 appears only as the terminal word, giving frequency $\approx 6.2\%$ rather than the uniform 25%. The 3-node fraction (classes 3+7) is 62.46%; the implied effective per-word termination probability is $0.6246/2 \approx 0.312$. The theoretical value is $1/4$ (Theorem 5.1); Lemma 4.1 proves $d_3 = d_7$ which makes the exact rate $1/4$.

Sentence Length Distribution

Statistical Tests

A χ^2 goodness-of-fit test on 15 length bins plus a tail bin, using $N = 10^5$ sentences:

Model	χ^2	p -value	Verdict
Naive $(1/2)^k$	1,022,029	≈ 0	Rejected
Theory $3^{k-1}/4^k$	12.9	0.61	Consistent

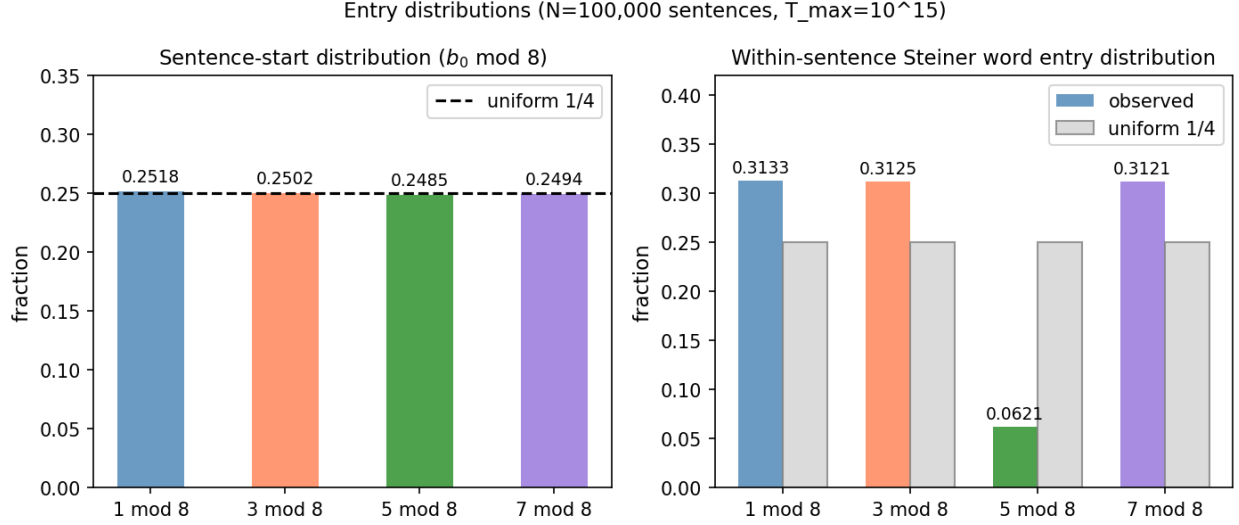


Figure 1: Left: sentence-start distribution ($b_0 \bmod 8$) against the uniform $1/4$ baseline — confirmed uniform (Table 1). Right: within-sentence Steiner word entry distribution against the uniform $1/4$ baseline. Class 5 is structurally suppressed to $\approx 6.2\%$ (it can only appear as the terminal word); classes 1, 3, 7 each carry $\approx 31\%$.

The naive model is rejected with overwhelming confidence. The theoretical formula $3^{k-1}/4^k$ is consistent with the data at all tested lengths.

References

- [1] J. Seymour, *A Regular Expression Language for the Collatz Graph* (Working Paper, Paper 33), 2026.
- [2] J. Seymour, *The $\frac{1}{3}(\frac{3}{4})^k$ Distribution of Steiner Sentence Lengths* (Working Paper, Paper 65), 2026.
- [3] J. Seymour, *2-adic Valuations of $3n + 1$ by Residue Class mod 8* (Working Paper, Paper 66), 2026.

k	Observed fraction	Naive $(1/2)^k$	Theory $3^{k-1}/4^k$
1	0.25180	0.50000	0.25000
2	0.18683	0.25000	0.18750
3	0.13931	0.12500	0.14063
4	0.10574	0.06250	0.10547
5	0.07844	0.03125	0.07910
6	0.05958	0.01563	0.05933
7	0.04490	0.00781	0.04449
8	0.03342	0.00391	0.03337
9	0.02490	0.00195	0.02503
10	0.01866	0.00098	0.01877

Table 3: Empirical Steiner sentence length distribution ($N = 10^5$, $T_{\max} = 10^{15}$) vs. the naive $(1/2)^k$ prediction and the theoretical $3^{k-1}/4^k$ (Theorem 5.1). The naive prediction overestimates $P(k=1)$ by a factor of 2 and decays far too fast for $k \geq 3$.

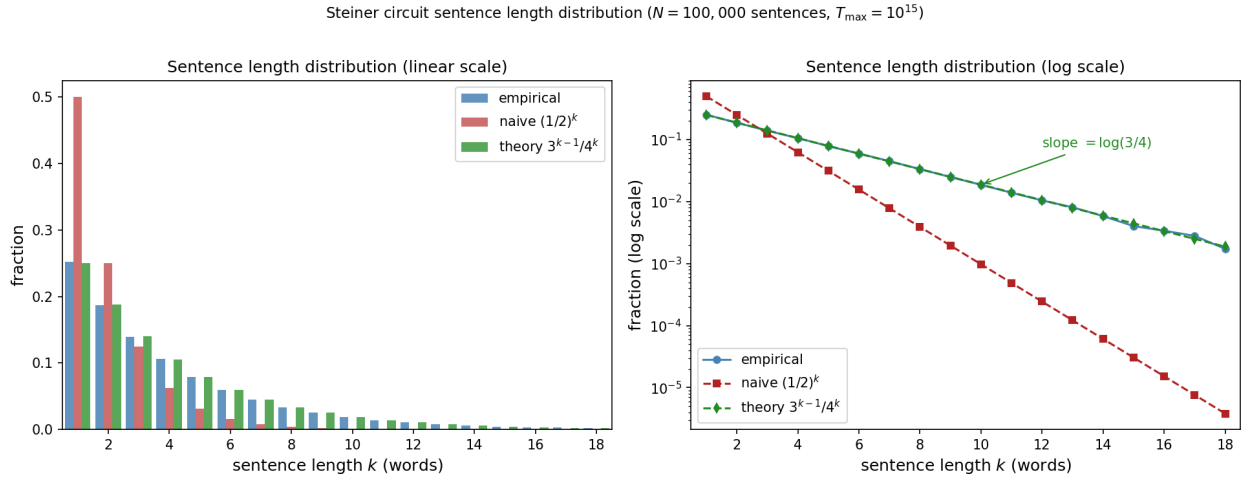


Figure 2: Steiner sentence length distribution ($N = 10^5$, $T_{\max} = 10^{15}$): empirical (blue bars / circles) vs. naive $(1/2)^k$ (red) and theoretical $3^{k-1}/4^k$ (green), on linear scale (left) and log scale (right). On the log scale, the empirical slope matches $\log(3/4)$ precisely — not $\log(1/2)$. The naive prediction is dramatically refuted at $k = 1$ (predicts 50%; observed 25%) and for $k \geq 3$ (predicts too-fast decay).